Qingyue Jiao

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U.S. Permanent Resident

Professional Summary

Computer Science Ph.D. candidate specializing in agentic AI, multimodal learning, and memory systems. Develops architectures and evaluation benchmarks for multimodal evidence integration, memory-augmented reasoning, and reliable decision-making.

Education

University of Notre Dame — Notre Dame, IN
Ph.D. Candidate in Computer Science and Engineering; Advisor: Prof. Yiyu Shi — *Sep 2024 – Present*

Columbia University — New York, NY
MS in Computer Science, GPA: 4.0 — *Sep 2022 – Dec 2023*

University of Michigan, Ann Arbor — Ann Arbor, MI
B.S. in Computer Science & Physics (Double) GPA: 3.88 — *Sep 2018 – May 2022*

Research Experience

University of Notre Dame — Notre Dame, IN
Ph.D. Candidate (Advisor: Prof. Yiyu Shi) — *Sep 2024 – Present*

MemEye: A Visual-Centric Evaluation Framework for Multimodal Agent Memory

- Designed a two-axis benchmark for **multimodal agent memory**, separating visual-detail retention from reasoning over changing states to enable targeted diagnosis of agent memory failures.
- Built a benchmark of 1,200 questions across eight scenarios and 12 evaluation categories using a construction pipeline with checks for answerability, visual dependence, and textual shortcuts.
- Evaluated 13 memory methods across four **vision-language model (VLM)** backbones, identifying lost visual details, stale retrieval, and state-tracking failures as bottlenecks in long-term agent reliability.

Hierarchical Memory for Radiology Report Generation with Rich Context

- Developed HM-RRG, combining **hierarchical memory** and **retrieval-augmented generation (RAG)** to ground chest X-ray reports in similar cases and longitudinal patient histories.
- Designed a **multimodal evidence integration mechanism** using image-conditioned memory compression and segment-level retrieval to combine cross-patient reports with longitudinal patient histories; fine-tuned BioMistral-7B with **LoRA** for evidence-grounded report generation.
- Improved CheXbert clinical F1 from 0.483 to 0.580 (+20.1% relative) over flat context concatenation on MIMIC-CXR, demonstrating the benefit of structured memory for clinical accuracy.

Agents' Last Doubt (SURE-Act): When to Act and When to Step Back in Tool-Using Agents

- Developed SURE-Act, a **decision controller** for frozen **tool-using LLM agents** that selects actions by balancing predicted task success, interaction cost, and irreversible-error risk.
- Built a **counterfactual rollout** pipeline with 3,115 action outcomes from 1,036 ToolSandbox decision points. Fine-tuned Qwen3-1.7B with **LoRA**, outcome regression, and **pairwise ranking** to predict action utility.
- Improved mean ToolSandbox Overall Score by 0.0216–0.0575 across three Qwen3 backbones on 89 unseen task families using learned action selection and an intervention stopping rule.

Seeing, Maintaining, and Learning: The Evolution of Multimodal Agent Memory

- Co-organized an 80-author collaboration and co-developed a multimodal agent memory taxonomy spanning representations, **adaptive management**, and **self-evolving memory policies**.
- Curated an open-source reference covering 296 memory architectures and 83 evaluation resources across five modality categories, supporting systematic comparison of memory designs and benchmark coverage.
- Analyzed trade-offs in evidence preservation, storage efficiency, and temporal consistency, translating findings into design guidance for memory writing, retrieval, updating, and forgetting.

Node-as-Agent: Graph Agentic Network

- Developed ReaGAN, a **training-free, LLM-powered Node-as-Agent** framework enabling autonomous planning over local aggregation and global retrieval for node classification.
- Designed a **local-global memory** mechanism that combines graph-neighbor evidence with semantically relevant nodes retrieved through RAG, extending evidence access beyond direct graph connections.
- Evaluated node classification on Cora, Citeseer, and Chameleon with a frozen Qwen2.5-14B model; achieved 84.95% accuracy on Cora without task-specific fine-tuning.

MediQ-GAN: Medical Image Generation from Limited Data

- Implemented a prototype-conditioned **hybrid GAN** combining classical and quantum-inspired modules for **synthetic data augmentation** of underrepresented medical image classes.
- Designed and ran augmentation experiments across three datasets and two classifier backbones; improved ViT-small accuracy on ISIC2019 from 72.49% to 82.60% (+10.11 percentage points).
- Analyzed generator **Jacobian spectra and effective rank** to diagnose mode collapse, showing broader latent-space utilization than five classical GAN baselines.

Publications

[1] **MemEye: A Visual-Centric Evaluation Framework for Multimodal Agent Memory**
Minghao Guo*, **Qingyue Jiao***, Zeru Shi*, et al.
*International Conference on Learning Representations (**ICLR 2027**)* (under review). arXiv:2605.15128.

[2] **Hierarchical Memory for Radiology Report Generation with Rich Context**
Qingyue Jiao, Jiahao Zheng, Jun Zhuang, Yiyu Shi.
*International Conference on Medical Image Computing and Computer-Assisted Intervention (**MICCAI 2026**).*

[3] **Node-as-Agent: Graph Agentic Network**
Minghao Guo, Xi Zhu, **Qingyue Jiao**, Xiujin Liu, Haochen Xue, Chong Zhang, Shuhang Lin, Jingyuan Huang, Ziyi Ye, Yongfeng Zhang.
SE4AgenticAI @ IEEE BigData 2026. arXiv:2508.00429.

[4] **Seeing, Maintaining, and Learning: The Evolution of Multi-modal Agent Memory**
Minghao Guo*, **Qingyue Jiao***, Xiujin Liu, et al.
*Transactions on Machine Learning Research (**TMLR 2026**)* (under review).

[5] **OpenPM: Auditable Point-in-Time Evaluation for LLM Portfolio-Management Agents**
Xinying Cai*, Minghao Guo*, Jiahe Liu, **Qingyue Jiao**, Jiaojiao Han, Bangwei Guo, Yitao Long, Yuxuan Chen, Bohan Wu, Ruochen Wen, Wenxi Dai, Dimitris N. Metaxas, Raymond Li.
ACM SIGKDD Explorations 2026.

[6] **Agentic Multimodal RAG: Roles, Decisions, and Evaluation**
Tianyu Yang, Wenliang Guo, Xiujin Liu, Shijian Deng, Minghao Guo, **Qingyue Jiao**, et al.
*Transactions on Machine Learning Research (**TMLR 2026**)* (under review).

[7] **MediQ-GAN: Quantum-Inspired GAN for High Resolution Medical Image Generation**
Qingyue Jiao, Yongcan Tang, Jun Zhuang, Jason Cong, Yiyu Shi.
arXiv:2506.21015, 2025.

[8] **A Comparison on Constrain Encoding Methods for Quantum Approximate Optimization Algorithm**
Yiwen Liu*, **Qingyue Jiao***, Yiyu Shi, Ke Wan, Shangjie Guo.
*IEEE/ACM International Conference on Computer-Aided Design (**ICCAD 2024**).*

[9] **Synergizing Quantum Techniques with Machine Learning for Advancing Drug Discovery Challenge**
Zhiding Liang, Zichang He, Yue Sun, Dylan Herman, **Qingyue Jiao**, et al.
Scientific Reports, 14:31216, 2024.

*Equal contribution.

Technical Skills

Programming & Systems: Python, C++, SQL, Git, Docker, Linux, CUDA, HPC

ML & Agents: PyTorch, Hugging Face, vLLM, LLM agents, vision-language models, agent memory systems, RAG, LoRA, benchmark design